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Pleasanton Math Tournament

2026
■ Pleasanton Math Tournament ■
Team Round
Problems 1–10

Team Captain ______________________________

Other Team Members ______________________________

School ______________________________

DO NOT BEGIN UNTIL YOU ARE INSTRUCTED TO DO SO.

This section of the competition consists of 10 problems which the team has 20 minutes to complete. Team members may work together in any way to solve the problems. Team members may talk to each other during this section of the competition. This round assumes the use of calculators, and calculations also may be done on scratch paper, but no other aids are allowed. All answers must be complete, legible and simplified to lowest terms. The team captain must record the team's official answers on his/her own competition booklet, which is the only booklet that will be scored. If the team completes the problems before time is called, use the remaining time to check your answers.

Total Correct	Scorer's Initials

AoPS Academy Pleasanton

Jane Street

WOLFRAM

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1. ________________

Shreyas, Athithan, Vatsal, Arnav, and Tiger took a math test. Their average score was 84. After Athithan's score is removed, the average score of the remaining students becomes 87. What was Athithan's test score?

2. ____________ hours

Rohan is planning a study schedule and is deciding between two tutors. Medha charges a \$10 down payment and then \$12 per hour of tutoring. Dalbert charges a \$40 down payment and then \$8 per hour of tutoring. What is the least positive integer number of hours for which hiring Dalbert costs less than hiring Medha?

3. ________________

Shreyas is at the origin of the coordinate plane, where he can only move up or right by one unit at a time. Let p be the amount of ways that he can get to the point (9,9). However, Shreyas wants to pass through his house, located at (3,7). Given that the number of ways for Shreyas to go to point (9,9) while going through his house is q, find $\frac{p}{q}$ to the nearest hundredth.

4. ________________

Medha rolls two fair six-sided dice, and Vatsal rolls one fair twelve-sided die. Medha's score is the sum of the two dice, and Vatsals score is the number shown on the twelve-sided die. What is the probability that Medha's score is greater than or equal to Vatsal's score?

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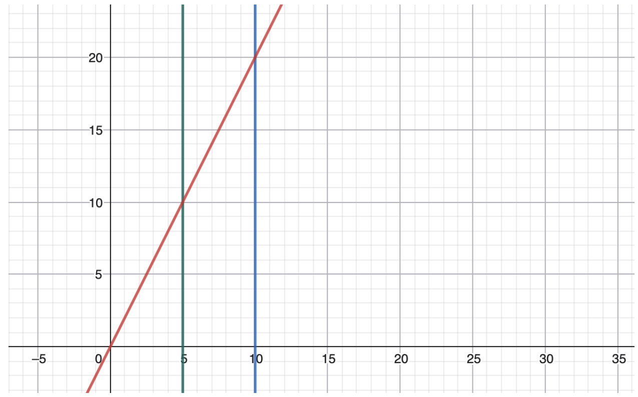
5. ____________ units^3

Consider the region bounded by the line $y = 2x$, the x-axis, and the vertical lines $x = 5$ and $x = 10$. This region is rotated one full revolution about the x-axis. Find the volume of the resulting solid, expressing your answer to the nearest hundredth.



6. ____________ units^2

Arnav has an infinitely thin rope of length 26. He wants to cut the rope somewhere so that after making an equilateral triangle with the first piece and a circle with the second piece, it will enclose as much area as possible. However, he must ensure that the triangle has a positive integer perimeter. What is the maximum area that Arnav can enclose with the rope following the above criteria? Express your answer as a decimal to the nearest hundredth.

7. ____________ meters

Let $ABCD$ be a rectangular fenced enclosure. A cow is somewhere on the plane of the enclosure. Its distances to points A, B, and C are 120, 117, and 90 meters respectively. What is the cow's distance from point D in meters to the nearest hundredth?

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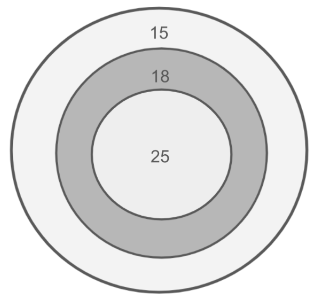
8. ____________ points

Taking inspiration from the Turkish athlete Yusuf Dikec, Vatsal decides to learn archery. However, he is not good at it, and every time he releases an arrow, it randomly lands somewhere on the target board. The target consists of 3 concentric circles with radii of 1, 2, and 3 respectively. If the arrow lands in the innermost circle, Vatsal gets 25 points. If it lands in the middle ring, he gets 18 points. If it lands in the outer ring, he gets 15 points. After shooting one arrow, how many points is Vatsal expected to earn? Express your answer as a common fraction.



9. ____________

Shreyas and his friend are playing a game with k marbles. On each turn, each player picks up a number of marbles ranging from 1 to 67. The goal of the game is to pick up the last marble. Shreyas goes second in this game. Assuming that Shreyas plays optimally, for how many positive integer values of k less than 2026 does he have a winning strategy?

10. ____________

Tiger connects the points on a 5 by 5 grid to make shapes. One day, he draws every possible non-degenerate rectangle with sides parallel to the grid lines and every possible non-degenerate triangle. If he draws r rectangles and t triangles, what is $\frac{r}{t}$ as a decimal to the nearest hundredth?

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